Chapter 9

Complex Engineering Problems (CEP) and Engineering Activities (CEA)

Not all problems in engineering are equal. Some can be solved by applying a known formula or following a familiar procedure. Others demand deeper thinking. They involve uncertainty, multiple constraints, conflicting requirements, and decisions that carry real consequences. It is within this space that Complex Engineering Problems and Engineering Activities emerge.

In Outcome-Based Education, especially within accredited engineering programs, students must go beyond routine exercises. They must learn to analyze ambiguity, design under constraints, evaluate alternatives, and justify decisions. CEP and CEA provide the framework to develop these capabilities systematically.

This chapter explores what makes a problem truly complex, how Engineering Activities differ from routine tasks, and how both should be embedded intentionally within courses. Understanding CEP and CEA is essential for designing meaningful learning experiences that prepare students for professional engineering practice.

9.1 Introduction

In Outcome-Based Education (OBE), **Complex Engineering Problems** (CEP) and **Complex Engineering Activities** (CEA) play a central role in demonstrating whether students can apply their knowledge and skills to real-world engineering contexts. According to BAETE, engineering graduates must be able not only to understand theory but also to solve complex problems and perform complex activities that involve uncertainty, multiple constraints, and professional judgment. This chapter explains CEP and CEA in a practical and faculty-friendly manner, with clear examples, mapping, justification, and common pitfalls.

9.2 Understanding CEP and CEA

Complex Engineering Problem (CEP) refers to a problem that cannot be solved using routine methods alone. It typically involves incomplete or conflicting requirements, multiple stakeholders, and the need to apply higher-order analysis and engineering judgment.

Complex Engineering Activity (CEA) refers to tasks or activities where students design, implement, test, or evaluate engineering solutions under realistic constraints. CEA focuses more on *doing*, while CEP focuses more on solving.

9.3 Key Characteristics (BAETE Perspective)

A task is considered **complex** when it involves one or more of the following:

- Requires in-depth analysis and engineering judgment

- Has multiple or conflicting constraints (technical, ethical, economic, societal)
- Cannot be solved by standard textbook procedures alone
- Requires integration of knowledge from different areas
- Involves uncertainty or incomplete information

Faculty should remember that **not every assignment or lab task qualifies as CEP or CEA**. Complexity must be purposefully designed.

9.4 When to Address Complex Engineering Problems (CEP)

Complex Engineering Problems should be addressed when students are expected to **analyze, model, and solve non-routine engineering problems** that cannot be handled using standard formulas or predefined procedures.

CEP is typically addressed in:

- Advanced level courses
- Design-oriented lab courses
- Capstone or Final Year Design Projects (FYDP)
- Research-based lab assignments
- Case studies involving real-world constraints

CEP focuses mainly on **problem understanding, analysis, and investigation.**

9.5 When to Address Complex Engineering Activities (CEA)

Complex Engineering Activities should be addressed when students are required to **design, implement, manage, or execute engineering work** in a realistic setting.

CEA is typically addressed in:

- Design projects and lab-based projects
- FYDP implementation and development phases
- Industry-based projects or internships
- Team-based engineering assignments

CEA focuses on **doing engineering work**, not just analyzing problems.

9.6 Program Outcomes Related to Complex Engineering Problems (CEP)

Complex Engineering Problems mainly contribute to the **PO1 to PO7**. These POs emphasize deep knowledge, analytical thinking, modeling, and research-based problem solving, which are core aspects of CEP.

Table 9.1: CEP attributes with descriptions.

Attribute	Description
P1: Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of **K3, K4, K5, K6** or **K8** which allows a fundamentals-based, first principles analytical approach.
P2: Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering and other issues.
P3: Depth of analysis required	Have no obvious solution and require abstract thinking and originality in analysis to formulate suitable models.
P4: Familiarity of issues	Involve infrequently encountered issues.
P5: Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering.
P6: Extent of stakeholder involvement and conflicting requirements	Involve diverse groups of stakeholders with widely varying needs.
P7: Interdependence	Are high-level problems including many component parts or sub-problems.

9.7 Program Outcomes Related to Complex Engineering Activities (CEA)

Complex Engineering Activities mainly contribute to the PO 10: Communication. This PO reflects professional engineering practice, teamwork, tools, and communication, which are central to CEA.

Table 9.2: CEA attributes with descriptions.

Attribute	Description
A1: Range of resources	Involve the use of diverse resources, including people, money, equipment, materials, information, and technologies.
A2: Level of interaction	Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering, or other issues.
A3: Innovation	Involve creative use of engineering principles and research-based knowledge in novel ways.
A4: Consequences for society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation.
A5: Familiarity	Can extend beyond previous experiences by applying principles-based approaches.

9.8 When Can a Problem Be Considered a Complex Engineering Problem?

A problem can be considered a **Complex Engineering Problem** when it satisfies **P1 (mandatory)** and **some or all of P2–P7** characteristics defined by BAETE (Table 11).

In simple terms, a problem is complex if:

- It requires **in-depth engineering knowledge** beyond basic formulas
- There is **no single or obvious solution**

- Multiple technical and non-technical factors are involved
- The problem involves uncertainty, trade-offs, or conflicting requirements
- Professional judgment and modeling are required

*If a problem can be solved by applying a standard method taught in a single class, it is **not** a CEP.*

9.9 How to Formulate a CEP for Students (Practical Guidance)

Faculty often ask how to convert a normal question into a CEP. The following steps help:

⇒ **Start with a real-world context**
Avoid purely theoretical problems. Use industry, social, or institutional scenarios.

⇒ **Introduce constraints**
Add limitations such as cost, security, ethics, performance, or sustainability.

⇒ **Avoid step-by-step instructions**
Let students decide the approach, assumptions, and tools.

⇒ **Encourage multiple valid solutions**
A CEP should not have a single correct answer.

⇒ **Link to higher Bloom's levels**
Use verbs such as *analyze, evaluate, design, and justify.*

Simple Transformation Example:

- Simple problem: "Normalize the given table."
- CEP version: "Analyze the database structure of an online shopping system and propose a normalized

design that minimizes redundancy while ensuring efficient query performance."

9.10 Common Mistakes in CEP and CEA Design

Faculty frequently make the following mistakes:

- Treating **routine numerical problems** as CEP
- Labeling **basic lab experiments** as CEA
- Providing overly detailed steps that eliminate complexity
- Mapping CEP/CEA to too many POs without justification
- Focusing only on implementation without analysis
- Not documenting CEP/CEA evidence for CQI and accreditation

Avoiding these mistakes ensures stronger alignment with BAETE expectations.

9.11 Best Practices for Faculty

- Clearly label CEP and CEA in course outlines and assessments
- Use rubrics aligned with CEP/CEA attributes
- Limit CEP/CEA tasks to key assessments (projects, case studies, FYDP)
- Collect and document evidence systematically
- Reflect on outcomes and close the CQI loop

9.12 Example of a Complex Engineering Problem

Course: Database Management Systems

Project Title:

Secure and Scalable Hospital Database System

Brief Project Description

Introduction:
Hospitals require reliable database systems to manage patient records, medical histories, billing information, and access control while ensuring data integrity and confidentiality.

Motivation:
The complexity of healthcare data, combined with ethical, security, and performance requirements, makes database system design a challenging engineering problem that goes beyond routine classroom exercises.

Expected Outcome:

Students analyze real-world requirements and propose a database design that ensures normalization, security, scalability, and ethical data handling.

CEP Attribute Mapping and Justification

Table 9.3: CEP mapping with justification

CEP Attribute	Justification
P1: Depth of knowledge required	Requires in-depth DBMS knowledge including ER modeling, normalization, and transaction concepts (K3, K4, K5).
P2: Range of conflicting requirements	Balances security vs. accessibility, normalization vs. performance, and usability vs. control.
P3: Depth of analysis required	No direct solution exists; students must analyze requirements and formulate suitable models.
P7: Interdependence	Design decisions in schema, security, and performance are interrelated and affect each other.

According to BAETE, a CEP does **not need to satisfy all attributes**. It is sufficient to demonstrate **strong alignment with selected attributes**. For this DBMS project, the following CEP attributes match **at best**:

Based on the justification given in the above table we can see that the problem clearly qualifies as a Complex Engineering Problem as defined by BAETE.

Table 9.4: Knowledge Profile Level's Justification

Knowledge Level	Justification
K3	Students must apply structured DBMS theory such as entity–relationship modeling, functional dependencies, normalization (1NF–BCNF), and integrity constraints to design a valid hospital database schema.

K4	Students analyze real-world hospital requirements including patient data flow, concurrent access, transaction management, and role-based access control. They must select appropriate mechanisms to ensure consistency, isolation, and confidentiality.
K5	The project requires advanced understanding of encryption, authentication, authorization, indexing strategies, performance optimization, and compliance with ethical handling of sensitive healthcare data.

Although the project includes design and implementation components, **Complex Engineering Activity (CEA) mapping is not mandatory for this course** for the following reasons:

- The defined **Course Outcomes (COs)** of the DBMS course are mapped to **PO-1 to PO-8 only**, and **no CO is mapped to PO-10.**
- BAETE expects **CEA evidence primarily where program outcomes related to complex activities are targeted.**
- In this course, the focus is on **problem analysis, modeling, design thinking, and ethical considerations,** which fall under **CEP**, not full-scale engineering activities.
- Routine implementation tasks in DBMS labs, while important, **do not independently satisfy the intent of Complex Engineering Activities** as defined by BAETE.

For this DBMS course, documenting a **well-justified CEP is sufficient** to meet BAETE expectations. CEA documentation is **not required** unless Course Outcomes explicitly target Program Outcomes associated with complex engineering activities. This approach ensures accurate alignment, avoids over-claiming, and strengthens accreditation compliance.

If, one day, we decide to map one of the DBMS course COs to **PO-10**, then introducing a **Complex Engineering Activity**

(CEA) will no longer be optional, it will be unavoidable. At that point, the students will not only *analyze* problems but will also be officially recognized for all the things they are already doing: designing schemas, writing SQL, testing systems, fixing errors, and documenting their work. As a DBMS teacher, it is easy to realize that students naturally perform many activities listed under the CEA attributes, even if we do not label them as such. And thankfully, **CQI exists for exactly this reason**. At the end of the semester, I can calmly write in the course report, "Based on observation and improvement needs, a new CO mapped to PO-10 is proposed." With that single line, the same project suddenly grows up and proudly qualifies as a CEA. Now, let's map the same project again, this time wearing the official CEA badge.

Table 9.5: CEA Attribute mapping with justifications.

CEA Attribute	Justification in the Given Activity
A1: Range of resources	Uses database software, hardware resources, SQL tools, documentation, teamwork, and time management.
A2: Level of interaction	Requires resolving conflicts between design efficiency, security requirements, and system performance.
A3: Innovation	Encourages creative database design choices and customized access-control strategies.
A4: Consequences for society and environment	Poor design may affect patient safety, data privacy, and ethical responsibility in healthcare systems.
A5: Familiarity	Extends beyond routine lab tasks by applying DBMS principles to a realistic and unfamiliar healthcare scenario.

9.13 Summary

Complex Engineering Problems focus on **analysis and problem solving**, while Complex Engineering Activities focus on **design, implementation, and professional practice**. Both are essential for Outcome Based Education and BAETE accreditation. Faculty members should consciously identify where CEP and CEA are addressed and document appropriate evidence in courses and projects.